

● EASY

● ALGEBRA

● ANSWER KEY

Linear functions

03

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**C**

C. $f(x) = 7x + 5$

Choice C is correct. An equation that defines a linear function f can be written in the form $f(x) = mx + b$, where m is the slope of the graph of $y = f(x)$ in the xy -plane and $(0, b)$ is the y -intercept of the graph. It's given that for the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 7. Therefore, $m = 7$. It's also given that the graph of $y = f(x)$ in the xy -plane passes through the point $(0, 5)$. Therefore, the y -intercept of the graph is $(0, 5)$, and it follows that $b = 5$. Substituting 7 for m and 5 for b in the equation $f(x) = mx + b$ yields $f(x) = 7x + 5$.

Choice A is incorrect. This equation defines a function whose graph has a slope of 5, not 7, and passes through the point $(0, 0)$, not $(0, 5)$.

Choice B is incorrect. This equation defines a function whose graph has a slope of 35, not 7, and passes through the point $(0, 0)$, not $(0, 5)$.

Choice D is incorrect. This equation defines a function whose graph has a slope of 12, not 7.

Q2**C**

C. Marisol drove at an average speed of about 45 miles per hour.

Choice C is correct. It's given that d is the distance, in miles, Marisol traveled after driving for t hours. Therefore, 45 represents the distance in miles traveled per hour, which is the speed she drove in miles per hour.

Choice A is incorrect and may result from misidentifying speed as the number of trips. Choice B is incorrect and may result from misidentifying speed as the total distance. Choice D is incorrect and may result from misidentifying the speed as the time, in hours.

Q3**A****A.** 2

Choice A is correct. Substituting 63 for $f(x)$ in the given function yields $63 = 9(2x + 3)$. Dividing both sides of this equation by 9 yields $7 = 2x + 3$. Subtracting 3 from both sides of this equation yields $4 = 2x$. Dividing both sides of this equation by 2 yields $2 = x$. Therefore, $f(x) = 63$ when the value of x is 2.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**B****B.** 37

Choice B is correct. It's given that the function f is defined by $fx = 4x - 3$. Substituting 10 for x in the given function yields $f10 = 4(10) - 3$, which is equivalent to $f10 = 40 - 3$, or $f10 = 37$. Therefore, the value of $f10$ is 37.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $f10$ for the function $fx = 4x$, not $fx = 4x - 3$.

Choice D is incorrect. This is the value of $f10$ for the function $fx = 4x + 3$, not $fx = 4x - 3$.

Q5**B****B. 6**

Choice B is correct. It's given that the equation $d = 16 - \frac{x}{30}$ gives the estimated amount of diesel d , in gallons, that remains in the gas tank of the truck after being driven x miles. Substituting 300 for x in the given equation yields $d = 16 - \frac{300}{30}$, which is equivalent to $d = 16 - 10$, or $d = 6$. Therefore, the estimated amount of diesel that remains in the gas tank of the truck when $x = 300$ is 6 gallons.

Choice A is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 480$, not when $x = 300$.

Choice C is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 60$, not when $x = 300$.

Choice D is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 0$, not when $x = 300$.

Q6**C****C. 80**

Choice C is correct. It's given that the function f is defined by $fx = 25x + 30$. Substituting 2 for x in this equation yields $f2 = 25(2) + 30$, which is equivalent to $f2 = 50 + 30$, or $f2 = 80$. Therefore, the value of fx is 80 when $x = 2$.

Choice A is incorrect. This is the value of 252, not $25(2) + 30$.

Choice B is incorrect. This is the value of $25 + 2 + 30$, not $25(2) + 30$.

Choice D is incorrect. This is the value of $25 + 30(2)$, not $25(2) + 30$.

Q7**B****B.** \$ 4.25

Choice B is correct. It's given that Josie purchased a pass to tour a nature preserve and 10 identical stickers for \$ 62.00 total. It's also given that the pass cost \$ 19.50. Let x represent the cost, in dollars, of one sticker. This relationship can be represented by the equation $19.50 + 10x = 62$. Subtracting 19.50 from both sides of this equation yields $10x = 42.50$. Dividing both sides of this equation by 10 yields $x = 4.25$. Therefore, the cost of one sticker was \$ 4.25.

Choice A is incorrect. This is the cost of one sticker if Josie purchased 10 identical stickers for \$ 19.50, not a pass and 10 identical stickers for \$ 62.00.

Choice C is incorrect. This is the cost of one sticker if Josie purchased 10 identical stickers for \$ 62.00, not a pass and 10 identical stickers for \$ 62.00.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**A****A.** \$2.00

Choice A is correct. The cost of each additional mile traveled is represented by the slope of the given line. The slope of the line can be calculated by identifying two points on the line and then calculating the ratio of the change in y to the change in x between the two points. Using the points $(1,5)$ and $(2,7)$, the slope is equal to $\frac{7-5}{2-1}$, or 2. Therefore, the cost for each additional mile traveled of the cab ride is \$2.00.

Choice B is incorrect and may result from calculating the slope of the line that passes through the points $(5,13)$ and $(0,0)$. However, $(0,0)$ does not lie on the line shown. Choice C is incorrect. This is the y -coordinate of the y -intercept of the graph and represents the flat fee for a cab ride before the charge for any miles traveled is added. Choice D is incorrect. This value represents the total cost of a 1-mile cab ride.

Q9**30**

The correct answer is 30. The value of fx when $x = 4$ can be found by substituting 4 for x in the given equation $fx = 7x + 2$. This yields $f4 = 7(4) + 2$, or $f4 = 30$. Therefore, when $x = 4$, the value of fx is 30.

Q10**B**

B. 38

Choice B is correct. It's given that function f is defined by $fx = 80 - 6x$. The value of $f7$ can be found by substituting 7 for x in the given function, which yields $f7 = 80 - 6(7)$, or $f7 = 80 - 42$, which is equivalent to $f7 = 38$. Therefore, the value of $f7$ is 38.

Choice A is incorrect. This is the value of $80 - 6(7)$, not $80 - 67$.

Choice C is incorrect. This is the value of $80 - 6(1)$, not $80 - 67$.

Choice D is incorrect. This is the value of $80 - 6 + 7$, not $80 - 67$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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